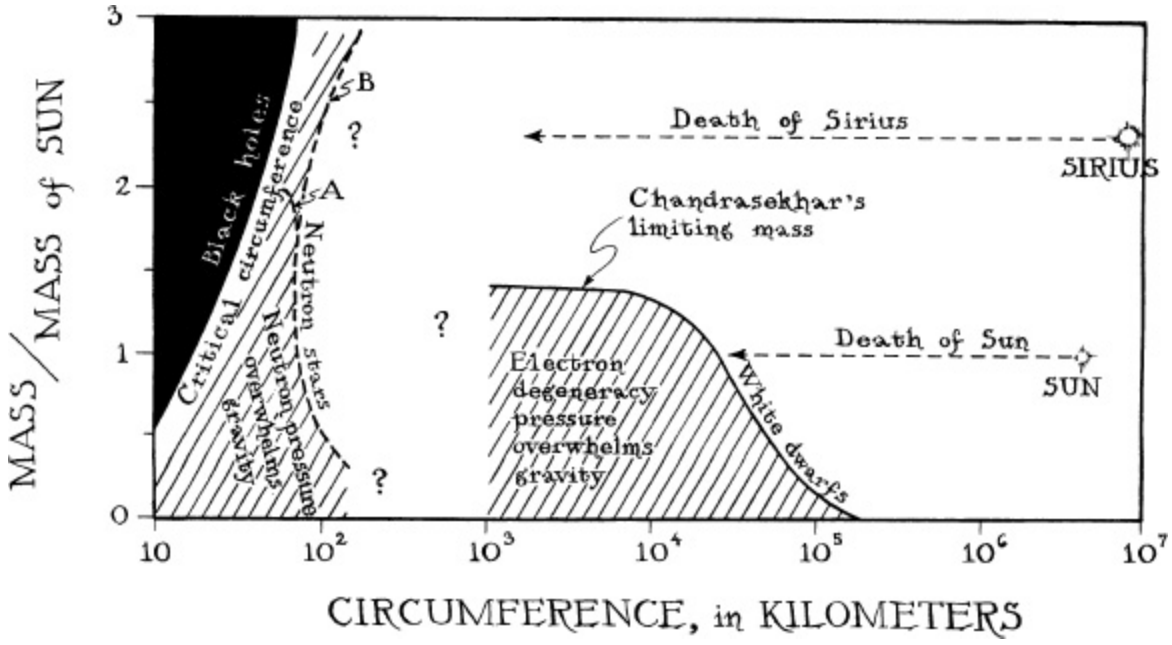


5.3 The ultimate fate of a star more massive than the Chandrasekhar limit of 1.4 Suns depends on how massive neutron stars can be. If they can be arbitrarily massive (curve *B*), then a star such as Sirius, when it dies, can only implode to form a neutron star; it cannot form a black hole. If there is an upper mass limit for neutron stars (as on curve *A*), then a massive dying star can become neither a white dwarf nor a neutron star; and unless there is some other graveyard available, it will die a black-hole death.

Suppose, instead, that the neutron-star curve does not extend upward in Figure 5.3 to large masses, but bends over in the manner of the hypothetical curve marked *A.* This will mean that there is a maximum mass that any neutron star can have, analogous to the Chandrasekhar limit of 1.4 Suns for white dwarfs. As for white dwarfs, so also for neutron stars, the existence of a maximum mass would herald a momentous fact: In a star more massive than the maximum, gravity will completely overwhelm the neutron pressure. Therefore, when so massive a star dies, it must either eject enough mass to bring it below the maximum, or else it will shrink inexorably, under gravity's pull, right past the neutron-star curve, and then—if there are no other possible stellar graveyards, nothing but white dwarfs, neutron stars, and black holes-it will continue shrinking until it forms a black hole.

Thus, the central question, the question that holds the key to the ultimate fate of massive stars, is this: *How massive can a neutron star be?* If it can be very massive, more massive than any normal star, then black holes can never form in the real Universe. If there is a maximum possible mass for neutron stars, and that maximum is not too large, then black holes